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Sensorless Contact-Force Estimation on Low-DoF, Velocity-Actuated Manipulators: A Quantitative Design Framework

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ABSTRACT Compliant physical interaction on low-cost manipulators must work without a force/torque sensor, whose price exceeds that of the arm, and with servos commanded in velocity rather than torque. That a residual observer cannot resolve a full wrench from fewer than six joints, and that a point-contact model reduces the unknown to three dimensions, is established; how much force is lost, in which directions, and at which postures is not. This paper supplies that quantification for a 4-DoF arm. The force block of the wrench projector predicts which directions are lost, and over 2197 configurations the six-dimensional inversion discards 50.5 % of an applied unit force on average while still correlating with ground truth at 0.997, so a validation that looks convincing can hide a large gain error. The magnitude of that loss depends on the characteristic length that makes the wrench norm dimensionally homogeneous, a choice the unweighted pseudoinverse makes silently; the directions lost do not, and the point-contact inversion admits no such choice. Velocity-level actuation, usually regarded as an obstacle, instead confines the dynamic model to the estimator: the residual is invariant to the inner servo loop, whose gain therefore need not be identified, and a kinematic controller reproduces the compliant behavior at one twenty-fifth of the cost. All results are obtained in simulation, against an independently written dynamic implementation that serves as the plant and establishes implementation correctness, not physical realism. The hardware experiment that would test the directional prediction, the conditioning law, and the viability of torque inferred from motor current is specified but not run.

INDEX TERMS Compliant control, force estimation, low-cost robotics, model predictive control, physical human-robot interaction, sensorless sensing, velocity control.

I. INTRODUCTION

COMPLIANT physical interaction, in which a manipulator yields to an external push and recovers its task once released, underpins hand guiding, contact-rich assembly, and safe operation in shared workspaces. The canonical implementation measures the interaction wrench with a wrist force/torque (F/T) sensor and closes an impedance or admittance law around it [1], [2].

On research-grade manipulators this is unremarkable. On arms assembled from commercial smart servos, a class that increasingly serves education, small-scale automation, and mobile manipulation research, it is not viable. A six-axis F/T sensor typically costs more than the entire mechanism, and mounting one consumes payload and structural stiffness that

a lightweight arm cannot spare. The economic argument that motivates a low-cost arm is precisely the argument against instrumenting it.

The question this paper addresses is therefore a design question rather than a feasibility one: given a low-DoF arm driven by velocity-controlled smart servos, what contact force is identifiable, with what accuracy, and how can it be used for compliant interaction?

A. PRIOR RESULTS AND THE REMAINING GAP

The established sensorless alternative is a momentum or residual observer, which compares measured joint accelerations and actuator torques against a dynamic model and attributes the mismatch to an external wrench [3]–[5]. The

approach is mature and requires no hardware beyond what a servo already reports; [6] surveys its detection, isolation, and identification variants.

The residual lives in joint space and is mapped to a Cartesian wrench through the transpose Jacobian pseudoinverse. This step recovers six unknowns from n measurements. That it therefore requires $n \geq 6$, and that restricting the unknown to a force at a known point reduces the task dimension to three, is stated explicitly by Magrini *et al.* [5], who also demonstrate on a seven-joint arm that attempting the six-dimensional inversion from too few informative residual components returns a wrong force together with a moment that should be zero. The dimensional argument is not in dispute and is not claimed here.

What that argument does not supply is anything a designer can act on. It does not say how much force is lost, and the answer is not a single number: it depends on posture and on the direction of the push, and the deficit reappears as a spurious moment rather than as a visible failure. At the task configuration $\mathbf{q} = [0, 1.2, -0.9, 0.4]$ rad of the arm studied here, a 6 N push normal to the plane of the arm is returned by the six-dimensional inversion as 0.264 N of force and $1.230 \text{ N} \cdot \text{m}$ of moment: 96 % of the force is reassigned. Two force directions at that same posture pass through unaltered. A statement about ranks cannot distinguish these cases. Nor is the arm the limitation: the same push produces $1.2867 \text{ N} \cdot \text{m}$ at the base joint, so the information is present in the measurement and the estimator discards it.

The failure is also silent. Section VII-A reports a compliant loop in which a push normal to the plane of the arm returns 83.4 % error, wrong by a factor of six. In the independent-engine check of Section VII-B the same inversion tracks the true force at correlation 0.997 while carrying a 2.227 N error, because correlation measures shape and the misattribution is a gain error. A validation that would ordinarily be accepted therefore certifies an estimator wrong by a factor of four.

B. VELOCITY-LEVEL ACTUATION

The second half of the design question is the actuator. Smart servos of this class are commanded in velocity, and the residual observer is formulated in torque. The apparent mismatch is often taken to mean that sensorless estimation is unavailable on such hardware, or that the servo loop must first be identified.

Neither is the case, and the reason is that the servo exposes a third signal besides position and velocity: the effort it is applying, reported as motor current. That channel is the quantity the residual requires, and it takes the place of the force/torque sensor throughout this work. Since the residual consumes the applied torque as measured rather than as modeled, the internal velocity loop may remain a black box.

Section V works out what this changes for the chain from motor current to commanded joint velocity, and the consequences run in the favorable direction: the interaction signal migrates from the acceleration to the actuator torque, the dynamic model is confined to the estimator, and the actuator

TABLE 1. Residual-based interaction sensing and the analyses closest to it: operating regime of representative approaches. A dash marks a quantity the cited work does not report, not one it reports as zero.

	joints	actuation	inversion	error reported
[3]	≥ 6	torque	six-dimensional	—
[4]	7	torque	six-dimensional	—
[5]	7	torque	point contact	hardware, one pose
[6]	≥ 6	torque	six-dimensional	—
[7]	7, 3	—	—	index, per posture
[8]	6	torque	six-dimensional	index, over workspace
this work	4	velocity	point contact	force, over workspace

saturates below the force range of manual guidance, so the arm yields physically without any control effort.

C. CONTRIBUTIONS

As shown in Table 1, residual-based interaction sensing has been developed for torque-actuated manipulators with six or more joints, where the wrench inversion is over-determined and the actuator accepts the quantity the controller computes. Neither condition holds on a low-cost smart-servo arm. To close this gap, this paper makes the following two contributions:

- We characterize the six-dimensional wrench inversion on an arm with fewer than six joints, quantitatively and geometrically. The force block of the wrench projector predicts which directions are lost, and over 2197 configurations the inversion discards 50.5 % of an applied unit force on average. The magnitude of that loss depends on the characteristic length that renders the wrench norm dimensionally homogeneous, a choice the unweighted pseudoinverse makes silently; the directions do not.
- We present an estimation and compliance architecture for low-DoF arms actuated in velocity, spanning the chain from servo current to commanded joint velocity. Because the applied torque is measured rather than modeled, the inner velocity loop never needs to be identified and the dynamic model is confined to the estimator; a kinematic controller reproduces the compliant behavior at one twenty-fifth of the cost.

The predictive controller of Section VI is an integration demonstrator, not an independent contribution: it shows that the estimate closes a compliant loop under the constraints of the platform, and its formulation is standard.

D. SCOPE AND ORGANIZATION

All results reported here are obtained in simulation, against a second and independently written dynamic implementation that agrees with the model to 10^{-10} and that also serves as the plant for the interaction experiments. Agreement at that level is evidence that two implementations of the same mechanics coincide, which is a check on correctness and not a substitute for physical validation: both share the rigid-body assumptions, and no simulator can adjudicate whether current is a usable proxy for torque on a particular servo.

Hardware validation is not claimed, and the accuracy claims should be read as conditional on it: the quantities most likely to degrade are precisely those the simulator cannot reproduce, namely motor current as a proxy for applied torque, gearbox friction and stiction, backlash, and the calibration of the torque constant. Section VIII states the limitations and specifies the experiment that would settle them. Section II positions the work, Section III describes the platform and model, Sections IV to VI develop the method, Section VII reports experiments, and Section IX concludes.

II. RELATED WORK

A. SENSORLESS INTERACTION SENSING

Residual-based external wrench estimation originates in collision detection. De Luca and Mattone introduced the sensorless residual for hybrid force/motion control [3], and the method was extended to safe reaction on lightweight arms [4]. Magrini *et al.* developed the virtual force sensor, which for redundant arms estimates the contact location along the structure in addition to the force [5], and the estimate has since been closed into hybrid force/velocity control [9] and into collaborative polishing [10]. A comprehensive account of detection, isolation, and identification appears in [6], and the operational-space formulation [11] supplies the mapping between joint and task quantities that all of these share. Learned corrections have also been applied to the same inversion, improving wrench estimates on serial arms in static tasks and behaving better near singular configurations [12].

The closest prior work is [5], and the relationship must be stated precisely, because the dimensional argument that motivates this paper is theirs. They establish that recovering both a contact force and a contact torque requires $\text{rank } \mathbf{J}_c = 6$, which holds only for $n \geq 6$; that restricting the contact to a force at a point reduces the task dimension to $m = 3$; that forces in $\mathcal{N}(\mathbf{J}_c^\top)$ are never recovered; and that $(\mathbf{J}_c^\top)^\# \mathbf{J}_c^\top \neq \mathbf{I}$ in general, so part of the contact force may not be identified. They also demonstrate the consequence on hardware: attempting the six-dimensional inversion where only four residual components are informative returns a wrong force together with a moment that should be zero. None of that is claimed here.

Three things separate the present work from that account, and they are matters of degree and regime rather than of discovery. First, their statement is qualitative and evaluated at a contact instance, whereas a designer choosing a posture, a task placement, or an arm needs the magnitude of the loss as a function of configuration and push direction. Section IV supplies that through the spectrum of $\mathbf{P}_{ff}$, which predicts the directional structure, and through a workspace sweep, which supplies its prevalence. Second, in their seven-joint setting the point-contact model is one modeling option that also enables localizing the contact along the structure, whereas for $n < 6$ it is the only route to a consistent force estimate, so the consequences of getting it wrong are structural rather than incidental. Third, their platform is torque-actuated. The interaction between residual-based estimation and velocity-level

actuation, which is the norm on low-cost hardware and which determines what must be identified before the estimator can run, is not addressed there and is the subject of Section V.

B. CAPABILITY ANALYSIS OF MANIPULATORS

Manipulability ellipsoids [13] and force polytopes [14] describe the set of velocities or forces a manipulator can *produce*, and are standard tools for posture optimization. The dual question, which external contacts a manipulator can *perceive* through its own sensing, has been formalized by Wong and Suleiman as sensor observability analysis [7]. They map the axes of distributed directional sensors into the task frame, obtain a per-configuration index and ellipsoid of sensing quality, and show on a seven-joint arm that certain postures leave interaction forces along a Cartesian axis unobserved while a ground-truth force sensor registers them. They also note the connection to $\mathcal{N}(\mathbf{J}^\top)$, and use the index prescriptively, maximizing it in the null space or by quadratic programming to avoid blind configurations.

That work and the present one answer different questions about the same geometry, and the distinction is worth stating. Their index is a normalized measure of alignment between sensor axes and task axes; it reports whether a direction is observed, not how many newtons of an applied force are lost, nor where the lost part is deposited. The mechanism studied here is the second one: under the six-dimensional inversion the unobserved component does not vanish, it reappears as a spurious moment, which is what makes the estimate wrong rather than merely incomplete. Their deficiency arises from sensor placement on arms that are not task-dimension deficient, whereas here it is structural in $n < 6$; and their analysis is local and prescriptive, whereas Section IV is concerned with the magnitude and prevalence of the loss across a workspace and with the contact model that removes it. The force polytope reappears in Section V-C in its classical role, where it sets the threshold between synthetic and physical compliance.

C. ESTIMATION AND COMPLIANCE ON COMMERCIAL ACTUATORS

A parallel line targets hardware without joint torque sensing, where the residual must be built from motor current. Cartesian contact forces and torques have been estimated from motor signals alone on seven-joint arms [15], external wrenches have been reconstructed from internal signals for fine manipulation [16] and with fast payload recalibration [17], and disturbance observers with learned friction models have been used on industrial arms [18]; direct teaching driven by current-inferred torque has been demonstrated on a surgical assistant [19]. On low-cost hardware the point-contact reduction is already the practical default: Yen *et al.* build a virtual force sensor from motor current on a sensorless arm and state plainly that the contact moment is disregarded and the force is referred to the end-effector [20]. What that literature does not supply, and what Section IV quantifies, is what the alternative costs when the arm cannot support it. Closest to the

hardware considered here, de Wolde *et al.* bring impedance control to a current-controlled arm without force or torque sensors, calibrating the actuator current-to-torque ratios and the joint friction from the robot model and gravity [21]. Their calibration runs in the opposite direction to the one used here: it converts a commanded joint torque into a current, so the current is an output and the external force is never estimated, which is precisely why impedance rather than admittance is the natural choice there. Their control law also retains the Cartesian inertia and Coriolis terms. The chain of Section V instead reads a *measured* current, infers the external joint torque and the contact force from it, and issues a joint velocity; the actuator closes its own loop underneath, and the dynamic model stays inside the observer. On the control side, admittance is the standard route to compliance on position-controlled hardware [22], [23], and a velocity interface has been reported to yield smoother compliant behavior than a position interface on an industrial arm [24].

D. PREDICTIVE CONTROL WITH A COUPLED POSE ERROR

Tracking a pose in SE(3) requires a scalar error penalizing translation and rotation jointly. The demonstrator of Section VI expresses the translational part in the frame of the rotational error through the inverse left Jacobian of SO(3) [25] rather than adding the two independently. Real-time rates are enabled by the real-time iteration scheme [26] and by embedded solver frameworks [27]–[29].

E. ESTABLISHED RESULTS

Several ingredients of this framework are established individually, and the contribution should not be read as larger than it is. That recovering a full wrench from the joint-torque residual requires at least six joints, and that restricting the unknown to a force at a known contact point reduces the task dimension to three, are both stated in [5]. That a posture can leave a task-space direction unobserved by joint torque sensing, and that this can be measured by a per-configuration index, is established in [7]. That Jacobian conditioning governs the accuracy of the reconstruction and can be used to select postures is demonstrated in [8]. Motor current has been used in place of joint torque sensing [15], and the point-contact reduction is already the practical default on low-cost arms [20].

What none of them reports is how much of an applied force the six-dimensional inversion discards, in which directions, and how often across a workspace; that this quantity is not scale-free, and which part of it is; that the loss is silent under a conventional validation; or the velocity-actuated regime. The 50.5 % figure is likewise not a general result: it belongs to this arm, this workspace, and one choice of the length that renders the wrench norm homogeneous. It carries weight because the analysis of Section IV explains and predicts it, and because the same construction can be evaluated for any other arm before one is built.

III. SYSTEM DESCRIPTION AND MODELING

A. PLATFORM

The manipulator has $n = 4$ revolute joints driven by Dynamixel MX-28R smart servos, with a rigid gripper treated as a fixed payload. Actuation limits are $|\tau_i| \leq 1.4 \text{ N} \cdot \text{m}$ and $|\dot{q}_i| \leq 5.969 \text{ rad/s}$. Each servo reports joint position and velocity, and its current measurement provides a proxy for the applied actuator torque. The servo accepts a velocity command; there is no torque command and no F/T sensor.

The first joint is a base yaw with vertical axis, and the remaining three are pitch joints whose axes are mutually parallel and orthogonal to it. This layout, a yaw base followed by a planar arm, is the standard low-cost architecture and is what determines the structure of the estimation problem.

B. RIGID-BODY DYNAMICS

The joint-space dynamics under an external wrench F applied at the end-effector are

$$M(q)\ddot{q} + h(q, \dot{q}) = \tau + J(q)^\top F, \quad (1)$$

with $q \in \mathbb{R}^4$ the joint angles, $\tau \in \mathbb{R}^4$ the actuator torques, and

$$J(q) = \begin{bmatrix} J_v(q) \\ J_\omega(q) \end{bmatrix} \in \mathbb{R}^{6 \times 4} \quad (2)$$

the geometric end-effector Jacobian partitioned into linear and angular blocks. The wrench is correspondingly $F = [f^\top m^\top]^\top$, with f the contact force and m the contact moment in the world frame. The inertia matrix is assembled from per-link geometric Jacobians,

$$M = \sum_i \left(m_i J_{v_i}^\top J_{v_i} + J_{\omega_i}^\top R_i I_i R_i^\top J_{\omega_i} \right) + D_a, \quad (3)$$

where D_a is the servo rotor armature, and h , collecting Coriolis, centrifugal, and gravitational terms, follows from the Lagrangian.

C. SENSING AND ACTUATION INTERFACE

The servo exposes three measurements and accepts one command, and the way they are used is the whole architecture, shown in Fig. 1.

- *Position* q and *velocity* $\dot{q}$, from the internal encoder. The acceleration $\ddot{q}$ is obtained by finite differences of $\dot{q}$.
- *Effort*, reported as motor current and converted to the applied actuator torque $\tau_{\text{act}} = K_t i$ through a per-joint constant K_t . *This third signal is what replaces the force/torque sensor.* It is the only channel through which the interaction reaches the controller, and Section V shows that under velocity-level actuation it carries essentially the entire interaction signal.
- *Command* u , a joint velocity setpoint. There is no torque command.

D. THE VELOCITY-ACTUATED PLANT

Since the actuator is commanded in velocity, the plant seen by the controller is not (1). Modeling the internal servo loop as a proportional velocity regulator,

$$\tau = k_v (u - \dot{q}), \quad (4)$$

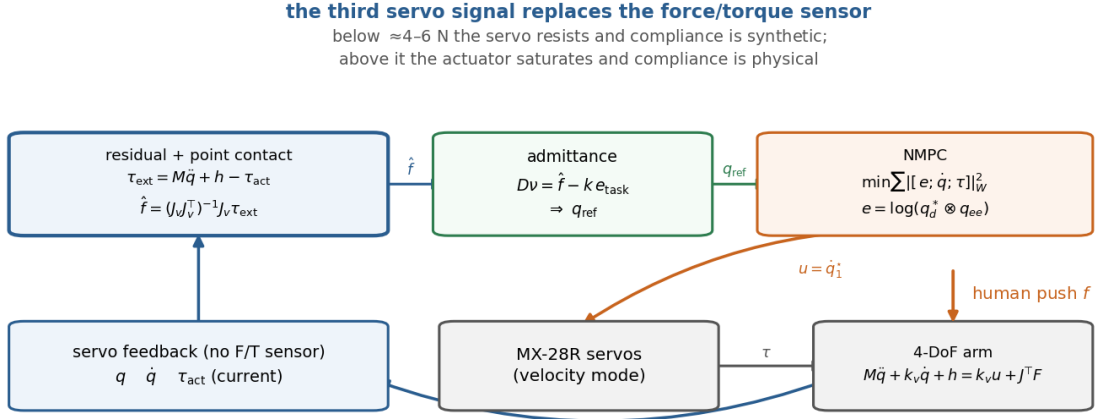


FIGURE 1. System architecture. The servo returns position, velocity, and effort; the effort channel takes the place of a force/torque sensor. The residual and the point-contact inversion produce $\hat{f}$, an admittance converts it into a displaced pose reference, and the predictive controller returns a joint velocity command, which is the only input the actuator accepts.

and substituting into (1) gives the closed-loop plant in compact form,

$$M(q)\ddot{q} + k_v\dot{q} + h(q, \dot{q}) = k_v u + J(q)^T F \quad (5)$$

which we verified against (1) to 4.3×10^{-14} over randomized states, gains, and wrenches. Equation (5) makes the role of the servo explicit: it contributes a damping term $k_v I$ and an input gain k_v , and as $k_v \rightarrow \infty$ it degenerates to the kinematic limit $\dot{q} = u$ in which the arm is a pure velocity source. Actuator saturation enters as $\|\tau\|_\infty \leq \tau_{\max}$ applied to (4), and Section V-C shows that this bound, rather than the loop gain, is what governs the interaction.

Remark 1 (The estimator does not use k_v). Equation (4) is written for analysis only. The residual (7) uses τ_{act} as measured through the effort channel, not as modeled, so k_v appears nowhere in the estimator. We verified that the recovered force is identical for $k_v \in \{1, 10, 100\}$. The internal servo loop may therefore remain a black box, and no system identification of it is required. This is the formal reason why velocity-level actuation does not obstruct sensorless estimation.

E. POSE ERROR

Tracking requires a scalar measure of pose error. We use a six-dimensional error whose rotational part is the axis-angle vector φ of the orientation error and whose translational part is expressed in the frame of that rotational error,

$$e = \begin{bmatrix} \varphi \\ \rho \end{bmatrix} \in \mathbb{R}^6, \quad \rho = J_\ell(\varphi)^{-1} t_{\text{err}}, \quad (6)$$

with J_ℓ the left Jacobian of $\text{SO}(3)$ [25] and t_{err} the translational error. The factor J_ℓ^{-1} distinguishes (6) from a decoupled error that adds position and orientation independently, and the two agree as $\|\varphi\| \rightarrow 0$. The choice matters only for the demonstrator, carries no weight in the contributions of this

paper, and costs nothing, both variants requiring 2.48 ms per solve.

F. MODEL VALIDATION

The symbolic model was built independently of the simulator and validated against MuJoCo [30] over eight configurations spanning the joint ranges. The maximum deviation is 8.8×10^{-11} on the inertia matrix M , 2.7×10^{-9} on the bias term h , 6.9×10^{-13} m on forward-kinematic position, 4×10^{-17} on the geometric Jacobian J , and zero to machine precision on forward-kinematic orientation. The agreement is at the numerical precision of the reference implementation, so modeling error is not a confounding factor in what follows. This matters because the central claim of this paper is that an observed estimation error belongs to the inversion step, and that attribution is credible only if the model itself is exact. It is a check that two implementations agree, not a measurement of the physical arm.

IV. CONTACT-MODEL-AWARE FORCE ESTIMATION

A. THE RESIDUAL AND THE ESTIMATION PROBLEM

Rearranging (1) gives the measurable residual

$$\tau_{\text{ext}} = M(q)\ddot{q} + h(q, \dot{q}) - \tau_{\text{act}} = J(q)^T F \in \mathbb{R}^4, \quad (7)$$

computed from joint measurements alone: q and $\dot{q}$ from encoders, $\ddot{q}$ by finite differences, and τ_{act} from the servo current. Everything the arm knows about the interaction is in these four numbers.

Problem 1. Given $\tau_{\text{ext}} \in \mathbb{R}^4$ from (7), recover the external contact force acting at the end-effector, and characterize the configurations in which recovery is possible and the accuracy attainable.

Problem 1 asks for a *force*, not a wrench. That distinction is the substance of this section: the number of unknowns

admitted into the inversion determines whether the problem is solvable, and the conventional choice admits too many.

B. THE SIX-DIMENSIONAL INVERSION

The conventional estimator treats the full wrench as unknown and solves $\mathbf{J}^\top \mathbf{F} = \boldsymbol{\tau}_{\text{ext}}$ for $\mathbf{F} \in \mathbb{R}^6$ from four equations. The system is under-determined and the minimum-norm solution is

$$\hat{\mathbf{F}}_6 = \mathbf{J}(\mathbf{J}^\top \mathbf{J})^{-1} \boldsymbol{\tau}_{\text{ext}} = \mathbf{P} \mathbf{F}, \quad \mathbf{P} = \mathbf{J}(\mathbf{J}^\top \mathbf{J})^{-1} \mathbf{J}^\top, \quad (8)$$

with $\mathbf{P}$ the orthogonal projector onto $\text{col} \mathbf{J} \subset \mathbb{R}^6$.

Proposition 1 (Force-to-moment misattribution). *For a pure contact force $\mathbf{F} = [\mathbf{f}^\top \mathbf{0}^\top]^\top$, the force returned by (8) is $\mathbf{P}_{\text{ff}} \mathbf{f}$, where $\mathbf{P}_{\text{ff}}$ is the leading 3×3 block of $\mathbf{P}$. In general $\|\mathbf{P}_{\text{ff}} \mathbf{f}\| < \|\mathbf{f}\|$, and the deficit appears as a nonzero moment in the lower block of $\hat{\mathbf{F}}_6$.*

The mechanism is misattribution, not loss of information. A pure force and a suitable force-moment pair produce identical joint torques, and minimum-norm selects the latter because it has smaller norm in $\mathbb{R}^6$. Both are consistent with the measurement; only one is realizable by the contact under consideration, and nothing in the formulation encodes which.

C. METRIC DEPENDENCE OF THE WRENCH NORM

Before that statement can be quantified, an implicit modeling decision has to be made explicit, because it sets the scale of every number that follows.

The minimum-norm criterion behind (8) minimizes $\|\mathbf{F}\|^2 = \|\mathbf{f}\|^2 + \|\mathbf{m}\|^2$ over the solution set. That sum is not dimensionally homogeneous: $\mathbf{f}$ is in newtons and $\mathbf{m}$ in newton-meters. Adding their squares presupposes a length. The unweighted pseudoinverse of $\mathbf{J}$ expressed in SI units, which is what the standard residual observer computes, therefore fixes that length at one meter without saying so. Equivalently, the top three rows of $\mathbf{J}$ carry units of meters and the bottom three are dimensionless, so $\text{col} \mathbf{J}$, and with it $\mathbf{P}$, changes if lengths are expressed in millimeters instead.

We make the choice a parameter. For a characteristic length $\ell_c > 0$, let

$$\mathbf{W}_{\ell_c} = \text{diag}(\mathbf{I}_3, \ell_c^{-1} \mathbf{I}_3), \quad \|\mathbf{F}\|_{\ell_c}^2 = \|\mathbf{f}\|^2 + \|\mathbf{m}\|^2 / \ell_c^2, \quad (9)$$

which is homogeneous in newtons. Minimizing $\|\mathbf{F}\|_{\ell_c}$ subject to $\mathbf{J}^\top \mathbf{F} = \boldsymbol{\tau}_{\text{ext}}$ gives

$$\hat{\mathbf{F}}_6(\ell_c) = \mathbf{W}_{\ell_c}^{-1} (\mathbf{J}^\top \mathbf{W}_{\ell_c}^{-1})^+ \boldsymbol{\tau}_{\text{ext}}, \quad (10)$$

and for a pure contact force the returned force is $\mathbf{M}(\ell_c) \mathbf{f}$, with $\mathbf{M}(\ell_c)$ the weighted counterpart of $\mathbf{P}_{\text{ff}}$. The unweighted inversion (8) is the case $\ell_c = 1$ m, and $\mathbf{M}(\ell_c)$ is symmetric for every ℓ_c .

Two further conventions must be stated for the quantities to be well posed. The moment is taken about the end-effector origin, the same point at which $\mathbf{J}$ is evaluated and at which the contact is assumed to act; expressing the wrench about another point changes $\mathbf{m}$ by $\mathbf{r} \times \mathbf{f}$ and therefore changes

$\mathbf{M}(\ell_c)$. And ℓ_c is a property of the arm, not of the contact: over the workspace sweep of Section IV-F the end-effector distance from the base has mean 0.233 m and maximum 0.365 m, so the arm's own scale is roughly 0.2 m, an order of magnitude below the meter that the unweighted convention imposes.

Two structural facts organize what follows. Writing $\mathbf{S}_{\ell_c} = \mathbf{W}_{\ell_c}^{-1}$, the minimizer of $\|\mathbf{F}\|_{\ell_c}$ subject to $\mathbf{J}^\top \mathbf{F} = \boldsymbol{\tau}_{\text{ext}}$ is $\hat{\mathbf{F}}_6(\ell_c) = \mathbf{W}_{\ell_c}^{-2} \mathbf{J}(\mathbf{J}^\top \mathbf{W}_{\ell_c}^{-2} \mathbf{J})^{-1} \boldsymbol{\tau}_{\text{ext}}$, so its range is $\text{col}(\mathbf{W}_{\ell_c}^{-2} \mathbf{J})$ and not $\mathbf{W}_{\ell_c}^{-1} \text{col} \mathbf{J}$. Substituting $\mathbf{W}_{\ell_c}^{-2} = \mathbf{S}_{\ell_c} \mathbf{S}_{\ell_c}$ gives

$$\mathbf{M}(\ell_c) = [\boldsymbol{\Pi}_{\ell_c}]_{1:3,1:3}, \quad \boldsymbol{\Pi}_{\ell_c} = \text{orthogonal projector onto } \text{col}(\mathbf{S}_{\ell_c} \mathbf{J}) \quad (11)$$

because the force block of $\mathbf{S}_{\ell_c}$ is the identity. Hence $\mathbf{M}(\ell_c)$ is symmetric and $\mathbf{0} \preceq \mathbf{M}(\ell_c) \preceq \mathbf{I}$ for every ℓ_c , and (8) is recovered at $\ell_c = 1$ m.

Proposition 2 (The exactly recovered set is metric-free). *For every $\ell_c > 0$, a pure contact force satisfies $\hat{\mathbf{F}}_6(\ell_c) = [\mathbf{f}^\top \mathbf{0}^\top]^\top$ if and only if $\mathbf{f} \in \mathbf{J}_v(\ker \mathbf{J}_\omega)$. The set does not depend on ℓ_c .*

Proof. Exact recovery holds when the true wrench lies in the range of the estimator, that is when $[\mathbf{f}^\top \mathbf{0}^\top]^\top \in \text{col}(\mathbf{W}_{\ell_c}^{-2} \mathbf{J})$. Since $\mathbf{W}_{\ell_c}^{-2} = \text{diag}(\mathbf{I}_3, \ell_c^2 \mathbf{I}_3)$, an element of that subspace reads $[(\mathbf{J}_v \mathbf{x})^\top \ell_c^2 (\mathbf{J}_\omega \mathbf{x})^\top]^\top$ for some $\mathbf{x} \in \mathbb{R}^4$. Its moment block vanishes if and only if $\mathbf{J}_\omega \mathbf{x} = \mathbf{0}$, a condition in which $\ell_c > 0$ cannot appear, and the force block is then $\mathbf{J}_v \mathbf{x}$. $\square$

The recovered set therefore consists of the forces produced by joint motions that generate no rotation, which is a statement about the mechanism and not about how newton-meters were scaled against newtons.

For the directional structure, invariance is *not* a general property of $\mathbf{M}(\ell_c)$: for an arbitrary Jacobian the eigenvectors do move with ℓ_c . It holds here, and it is worth proving rather than observing, because the reason is the architecture the paper is about.

Proposition 3 (Directional structure of a yaw-plus-parallel-pitch arm). *Let joint 1 rotate about the vertical $\mathbf{e}_z$ and let the axes of joints 2 to 4 be parallel to a common horizontal direction $\mathbf{a}$, the joint lines themselves being distinct. Let $\mathcal{P} = \mathbf{J}_v(\ker \mathbf{J}_\omega)$. If $\dim \mathcal{P} = 2$, then $\mathcal{P} = \mathbf{a}^\perp$ and*

$$\mathbf{M}(\ell_c) = \mathbf{I} - (1 - \lambda(\ell_c)) \mathbf{a} \mathbf{a}^\top, \quad \lambda(\ell_c) = \mathbf{a}^\top \mathbf{M}(\ell_c) \mathbf{a} \in [0, 1]. \quad (12)$$

The eigenvectors of $\mathbf{M}(\ell_c)$ are therefore independent of ℓ_c : the directions returned unaltered are the plane of the arm, and the least observable direction is the pitch axis $\mathbf{a}$. The eigenvalue $\lambda(\ell_c)$ does depend on ℓ_c , and with it every quantity built on the spectrum.

Proof. The columns of $\mathbf{J}_\omega$ are $\mathbf{e}_z, \mathbf{a}, \mathbf{a}, \mathbf{a}$, so $\ker \mathbf{J}_\omega = \{\mathbf{x} : x_1 = 0, x_2 + x_3 + x_4 = 0\}$, using that $\mathbf{e}_z$ and $\mathbf{a}$ are independent. The corresponding force directions are $\mathbf{J}_v \mathbf{x} = x_2 \mathbf{v}_2 + x_3 \mathbf{v}_3 + x_4 \mathbf{v}_4$ with $\sum x_i = 0$, where $\mathbf{v}_i = \mathbf{a} \times \mathbf{r}_i \perp \mathbf{a}$; hence $\mathcal{P} \subseteq \mathbf{a}^\perp$, and $\dim \mathcal{P} = 2$ forces equality. By Proposition 2 every $\mathbf{f} \in \mathcal{P}$ is recovered exactly, so $\mathbf{M}(\ell_c)$ acts as

TABLE 2. Sensitivity of the six-dimensional inversion to the characteristic length ℓ_c that makes the wrench norm homogeneous. Sweep of 2197 configurations, one random unit contact force at each.

ℓ_c [m]	force discarded [%]				configurations $\lambda_{\min} < 0.10$
	mean	median	p90	max	
0.05	12.8	5.4	37.1	99.1	6 %
0.10	22.9	15.7	54.7	99.5	11 %
0.15	30.2	24.9	63.3	99.6	17 %
0.20	35.4	32.2	69.8	99.6	24 %
0.233*	38.0	35.9	73.0	99.6	29 %
0.30	41.9	41.1	77.5	99.6	39 %
0.50	47.3	47.9	84.0	99.6	63 %
1.00†	50.5	51.1	89.2	99.6	100 %
2.00	51.4	52.2	91.1	99.6	100 %

†unweighted inversion in SI units, the standard practice. *mean end-effector reach of this arm.

the identity on $\mathbf{a}^\perp$ for every ℓ_c . Since $\mathbf{M}(\ell_c)$ is symmetric by (11), it leaves $(\mathbf{a}^\perp)^\perp = \text{span } \mathbf{a}$ invariant, which gives (12), and $\mathbf{0} \preceq \mathbf{M}(\ell_c) \preceq \mathbf{I}$ bounds λ . $\square$

The hypothesis $\dim \mathcal{P} = 2$ is what the numbers report as a spectrum $\{\lambda, 1, 1\}$: over the 2197 configurations of Section IV-F the two conditions agree in 100 % of cases and both hold in 92.3 %, the remainder being the singular configurations on the boundary of the grid, where the three pitch links align. Over that sweep and the nine values of ℓ_c in Table 2, restricted to the configurations satisfying the hypothesis, (12) is confirmed to 6×10^{-14} on $\mathcal{P}$ and 8×10^{-15} on $\mathbf{a}$, with $\|\mathbf{M}(\ell_c) - \mathbf{M}(\ell_c)^\top\|$ below 6×10^{-14} , and the eigenspaces agree with the $\ell_c = 1$ m case to 2.6×10^{-6} degrees.

Equation (12) gives the error law with no further assumption. For a unit force at angle θ to the pitch axis,

$$\|\mathbf{M}(\ell_c)\mathbf{f} - \mathbf{f}\| = (1 - \lambda(\ell_c)) |\mathbf{a}^\top \mathbf{f}| = (1 - \lambda(\ell_c)) |\cos \theta|. \quad (13)$$

The law is exact, its direction $\mathbf{a}$ is metric-free, and ℓ_c sets only its slope $1 - \lambda(\ell_c)$. Section IV-G measures both.

Table 2 quantifies the exposure. The mean discarded fraction moves from 12.8 % to 51.4 % across a plausible range of ℓ_c , and the count of configurations with a nearly unobservable direction moves from 6 % to 100 %, which is the most metric-sensitive statistic in this paper and is reported here rather than quoted as a headline figure. What does not move is the worst case, 99.6 % for every $\ell_c \geq 0.15$ m, and the identity of the direction in which it occurs.

Unless stated otherwise the remainder of the paper uses $\ell_c = 1$ m, referred to throughout as the *unweighted-SI baseline* rather than as a natural metric. It is not the natural scale of this arm; it is the metric that the estimator under examination actually implements. The object of study is standard practice, and the numbers reported are the ones standard practice produces. Readers who prefer the arm's own scale should read the $\ell_c = 0.233$ m row: the qualitative conclusion is unchanged and the magnitude drops from 50.5 % to 38.0 %.

Remark 2 (The point-contact inversion has no such parameter). Equation (15) below solves for $\mathbf{f} \in \mathbb{R}^3$ alone. Its norm is

already homogeneous, no length enters, and no weighting has to be chosen or declared. This is not a side benefit: restricting the unknown to the physically realizable set removes an arbitrary modeling decision that the six-dimensional formulation is forced to make and that is, in current practice, made silently.

At the task configuration $\mathbf{q}_{\text{task}} = [0, 1.2, -0.9, 0.4]$ rad, a 6 N push normal to the plane of the arm produces $\boldsymbol{\tau}_{\text{ext}} = [1.2867, 0, 0, 0]^\top \text{ N} \cdot \text{m}$. Equation (8) returns 0.264 N of force together with 1.230 N · m of moment, and the eigenvalues of $\mathbf{P}_{\text{ff}}$ there are $\{0.044, 1, 1\}$: two force directions pass unaltered while a third is almost entirely converted into a moment. At $\ell_c = 0.233$ m the same push returns 2.755 N and 0.696 N · m, with spectrum $\{0.459, 1, 1\}$. The misattribution is present in both; its severity is set by the metric, and the direction in which it occurs is not.

Remark 3 (Aggregate observability is uninformative). Since $\mathbf{P}$ is an orthogonal projector, $\text{tr } \mathbf{P} = \text{rank } \mathbf{J}$ identically, so for wrench directions drawn uniformly on the unit sphere of $\mathbb{R}^6$ the expected fraction returned by (8) is $\text{rank } \mathbf{J} / 6$ regardless of configuration. Any workspace-averaged observability index built on $\mathbf{P}$ is constant by construction and carries no information about posture. The structure lies in how the deficit is distributed across directions.

Remark 4 (Numerical care at singularities). At singular configurations $\mathbf{J}$ loses rank and (8) is undefined. Throughout this work $\mathbf{P}$ is computed from the thin singular value decomposition $\mathbf{J} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$ as $\mathbf{P} = \mathbf{U}_r\mathbf{U}_r^\top$ with r the numerical rank, which remains valid there.

D. THE POINT-CONTACT INVERSION

A human push at the end-effector transmits a force at a known point and no moment about that point. Imposing $\mathbf{m} = \mathbf{0}$ in (7) leaves

$$\mathbf{J}_v(\mathbf{q})^\top \mathbf{f} = \boldsymbol{\tau}_{\text{ext}}, \quad \mathbf{J}_v^\top \in \mathbb{R}^{4 \times 3}, \quad (14)$$

over-determined: four equations in three unknowns, with least-squares solution

$$\hat{\mathbf{f}}_3 = (\mathbf{J}_v \mathbf{J}_v^\top)^{-1} \mathbf{J}_v^\top \boldsymbol{\tau}_{\text{ext}}. \quad (15)$$

Proposition 4 (Exact recovery). $\hat{\mathbf{f}}_3 = \mathbf{f}$ for every pure contact force applied at the end-effector if and only if $\text{rank } \mathbf{J}_v = 3$. The result is independent of the rank deficit of $\mathbf{J}$ and does not involve $\mathbf{J}_\omega$.

Proof. By construction (14) is consistent, since $\boldsymbol{\tau}_{\text{ext}} = \mathbf{J}_v^\top \mathbf{f}$ for the true $\mathbf{f}$. A consistent least-squares problem attains zero residual, and its solution is unique precisely when $\mathbf{J}_v^\top$ has trivial kernel, that is when $\text{rank } \mathbf{J}_v = 3$. If $\text{rank } \mathbf{J}_v < 3$ the solution set is an affine subspace of dimension $3 - \text{rank } \mathbf{J}_v$ and (15) returns its minimum-norm element, which differs from $\mathbf{f}$ whenever $\mathbf{f}$ has a component in $\ker \mathbf{J}_v^\top$. The angular block $\mathbf{J}_\omega$ appears nowhere in (14), so the rank of $\mathbf{J}$ is irrelevant. $\square$

Proposition 4 moves the governing condition from $\mathbf{J} \in \mathbb{R}^{6 \times 4}$, which cannot attain rank 6, to $\mathbf{J}_v \in \mathbb{R}^{3 \times 4}$, which

generically attains rank 3. Four joints are not too few to sense a contact force at the end-effector; they are one more than the minimum required.

Remark 5 (Cost of the model assumption). *The point-contact model requires the contact location to be known. This holds for hand guiding at the end-effector and fails for contact along the links. If the contact additionally transmits a moment, for instance through a rigidly grasped object, the ambiguity of Proposition 1 returns and (8) becomes the honest choice, with the accompanying loss. Relaxing the assumption is an established line in its own right: the contact location can be recovered together with the force by constrained optimization [31] or by particle filtering over the robot surface [32], and it is what the virtual force sensor provides on redundant arms [5]. Extending the estimator in that direction is future work.*

E. CONDITIONING AS THE BINDING LIMITATION

With exactness settled, the practically relevant quantity is sensitivity to measurement noise. Perturbing the residual by $\delta\tau$ perturbs the estimate by

$$\|\delta\hat{f}_3\| \leq \frac{\|\delta\tau\|}{\sigma_{\min}(\mathbf{J}_v)}, \quad (16)$$

so $\sigma_{\min}(\mathbf{J}_v)$ bounds the amplification of residual error into force error and replaces observability as the quantity to map.

That Jacobian conditioning governs the accuracy of residual-based force estimation is established. Lu *et al.* bound the relative force error of a six-joint industrial robot by the condition number of $\mathbf{J}^\top$ times the relative residual error, map that condition number over half a million sampled configurations, and then optimize the task path and end-effector pose to lower it, thereby suppressing the effect of modeling error without reducing the error itself [8].

Three things differ here, and they follow from the regime rather than from the technique. Their reconstruction inverts $\mathbf{J}^\top$ directly, which presumes it is square and invertible; for $n < 6$ that inverse does not exist, and (16) is instead an absolute bound on the over-determined point-contact inverse (15), whose relevant quantity is the smallest singular value of $\mathbf{J}_v$ rather than a condition number of $\mathbf{J}$. Their workspace study maps the *index*; Section IV-F maps the *error*, and shows that its dominant component is the structural misattribution of Proposition 1, which survives at good conditioning and therefore cannot be removed by any posture choice. And where a six-joint arm executing a one-dimensional path has a redundant angle to spend on conditioning, a four-joint arm tracking a pose has none, so $\sigma_{\min}(\mathbf{J}_v)$ is used here as a gate that declines to report rather than as an objective to maximize. The bound is verified as a law on the simulated plant in Section VII-E.

F. WORKSPACE CHARACTERIZATION

We sweep the three pitch joints over their full ranges on a 13^3 grid, fixing the base yaw since it only rotates the

configuration in the world frame and leaves every quantity of interest invariant. This gives 2197 configurations, at each of which we compute the rank and singular values of $\mathbf{J}_v$ and apply a random unit contact force, recording the error of both estimators. Fig. 2 maps the outcome over a slice of that grid.

Two facts stand out. First, $\text{rank}(\mathbf{J}_v) = 3$ in 100 % of the grid, so by Proposition 4 the contact force is exactly recoverable everywhere. The numerical error of $\hat{f}_3$ on a unit force is 3.6×10^{-15} in the mean and never exceeds 1.4×10^{-12} , which is machine precision. That statement involves no metric and no length. What does limit the point-contact estimator is conditioning: $\sigma_{\min}(\mathbf{J}_v)$ has mean 0.0500 and median 0.0496 over the grid, falls to 0.0127 at the fifth percentile and to 7.6×10^{-5} at its worst, while $\text{cond}(\mathbf{J}_v)$ has median 6.7 and reaches 29.4 at the ninety-fifth percentile. Second, over the same set the six-dimensional inversion discards 50.5 % of the applied force on average under the unweighted-SI baseline, with median 51.1 % and a worst case of 99.6 %; under the arm's own characteristic length the mean falls to 38.0 % and the worst case stays where it was. Table 2 gives the full dependence; only the mean is metric-sensitive, and Propositions 2 and 3 identify which part of the conclusion that affects.

The geometric reason the six-dimensional inversion fails in a particular direction is worth stating, because it explains why the effect is systematic on this class of arm. The three pitch axes are parallel, so the torques those joints produce resist only loads in the plane of the arm. A force component normal to that plane reaches the actuators through a single path, the moment it exerts about the base yaw axis. A moment applied directly about that same axis produces the identical joint torque with a smaller norm in $\mathbb{R}^6$, so the minimum-norm criterion prefers it. The out-of-plane force direction is therefore exactly the one that the six-dimensional inversion converts into a moment, in every configuration of this architecture.

G. DIRECTIONAL CHARACTERIZATION

The preceding argument predicts that the error of the six-dimensional inversion is governed by the *direction* of the push and by nothing else. That prediction is falsifiable and we test it directly, since a claim of this kind cannot rest on the two push directions used in the closed-loop experiments of Section VII-A. We sweep the full sphere of push directions, 2000 uniformly drawn directions at each of six configurations spread over the workspace, and for each direction record the relative error of both inversions together with the alignment $|\cos \theta|$ between the push and the least observable eigenvector of $\mathbf{P}_{ff}$ at that pose.

Table 3 reports the result. Three observations follow. The median error of the six-dimensional inversion is between 47.7 and 49.5 % at every pose, and roughly half of all push directions incur an error above 50 %, so the failure is not confined to a special direction or a special posture. The point-contact inversion is exact to 4×10^{-13} % over all 12000 directions and all six poses. And the poses span $\sigma_{\min}(\mathbf{J}_v)$ from 0.0255 to 0.0802 without the six-dimensional error

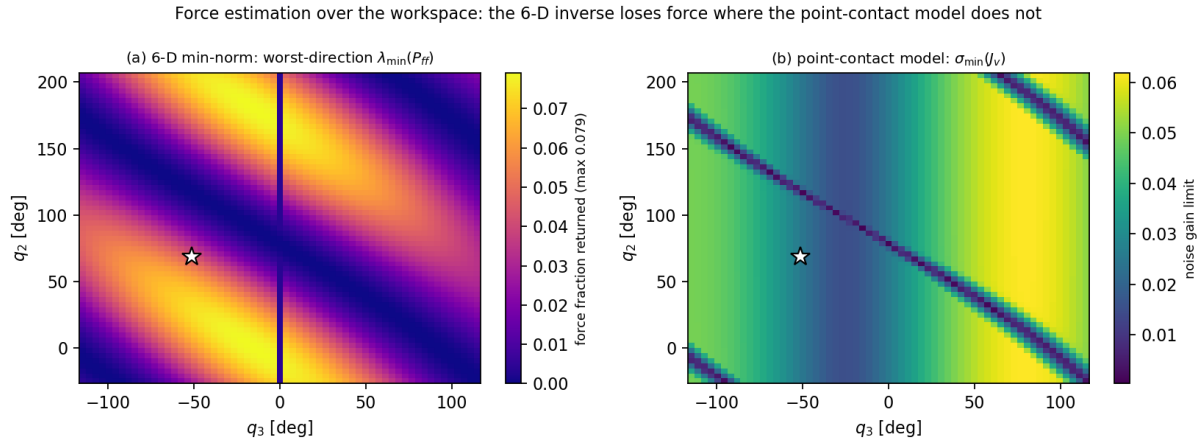


FIGURE 2. Force estimation over a slice of the workspace ($q_1 = 0$, $q_4 = 0.4$ rad). (a) Worst-direction eigenvalue of P_H at the unweighted-SI baseline, the fraction of a pure contact force that the six-dimensional inversion returns as force. Note the scale: it never exceeds 0.079 anywhere on the slice, so even the most favorable posture returns under a twelfth of a push along that direction, and the pattern is a property of the geometry rather than of any one configuration. This is the effect that has been misread as blindness of the mechanism. (b) $\sigma_{\min}(J_v)$, which bounds noise amplification for the point-contact inversion and is the quantity that actually limits it. The star marks the task configuration.

TABLE 3. Directional sweep. Relative force error over 2000 push directions at each of six configurations.

pose	$\sigma_{\min}(J_v)$	six-dimensional [%]			point contact max [%]
		median	p90	max	
P1	0.0255	47.7	85.7	95.6	4.1×10^{-13}
P2	0.0357	49.5	89.7	99.6	1.9×10^{-13}
P3	0.0597	49.1	86.3	96.5	1.9×10^{-13}
P4	0.0403	48.1	89.6	99.8	2.2×10^{-13}
P5	0.0802	48.8	85.5	94.9	1.5×10^{-13}
P6	0.0303	48.0	87.7	97.3	2.0×10^{-13}

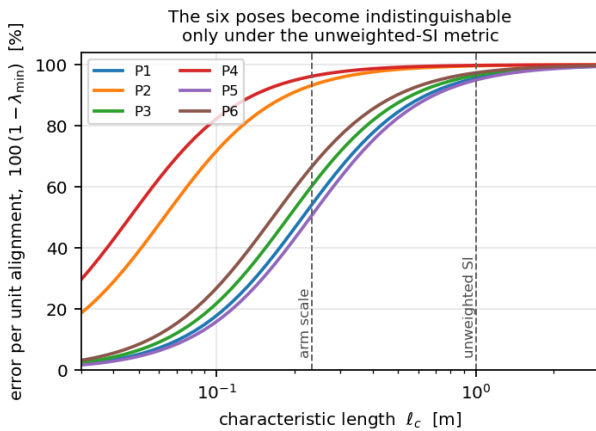


FIGURE 3. The slope of (13), $1 - \lambda_{\min}$, against the characteristic length, at the six configurations of the directional sweep. Each pose has its own slope at small ℓ_c and all six approach a common value as ℓ_c grows, so at the unweighted-SI baseline they are nearly indistinguishable while at the scale of the arm they are not. This is what the pooled correlation of 0.9993 measures and what it conceals. The direction that is lost is the same throughout, by Proposition 3, and it is only the slope that moves.

responding, confirming that its error is not a conditioning effect.

The pooled numbers confirm the mechanism quantitatively. Over the 12 000 directions the median six-dimensional error grows monotonically with alignment, from 9.8 % below $|\cos \theta| = 0.2$ to 94.8 % above 0.95, and the correlation between alignment and error is 0.9993. The geometric explanation is therefore not an interpretation placed on the numbers after the fact: it accounts for essentially all of the variance. Fig. 3 shows what that pooled figure conceals, and the next remark states it.

Remark 6 (Scope of the correlation). *The least observable direction is obtained from P_{ff} , that is from the same model that produces the error, so the correlation of 0.9993 should be read for what it is: evidence that a single scalar function of the push direction predicts the error, not an independent confirmation. What it excludes is that the error depends on magnitude, on posture, or on conditioning, all of which are varied across the sweep without affecting it. An independent test would identify the least observable direction empirically on hardware and check that it coincides with the one P_{ff} predicts.*

Remark 7 (The pooled correlation is flattered by the metric). *Two distinct statements are combined in the figure 0.9993, and only one of them is invariant. Within a single pose the relation is the algebraic identity (13), so the correlation is exactly unity at every pose and every ℓ_c ; this is verified to six decimals across the nine values of ℓ_c in Table 2. Pooling six poses adds a second effect, since each pose contributes a line of slope $1 - \lambda_{\min}$, and the pooled correlation measures how nearly those slopes agree. At $\ell_c = 1$ m every $\lambda_{\min}$ lies below 0.09, the slopes are nearly identical, and pooling costs almost nothing, which is why 0.9993 appears. At $\ell_c = 0.05$ m the same 12 000 directions give a pooled correlation of 0.4475 while every per-pose correlation is still exactly unity. The predictive claim is the per-pose identity; the pooled number should be read as a statement about this arm under the*

unweighted metric, not as the strength of the prediction.

V. ESTIMATION UNDER VELOCITY-LEVEL ACTUATION

The residual (7) is written in torque, while the actuator accepts velocity. This section shows that the mismatch is apparent, that velocity-level actuation in fact simplifies the estimation problem, and that it changes both the error budget and the nature of the compliance obtained.

Remark 1 already established the structural point: the residual uses τ_{act} as measured through the effort channel, so the servo gain k_v does not enter the estimator and the inner loop need not be identified. What must be known is the map from reported current to torque, which is a calibration rather than a system identification. The remainder of this section works out the quantitative consequences.

A. DISTRIBUTION OF THE INTERACTION SIGNAL

Simulating the plant (5) under a 6 N push at q_{task} , and holding $u = 0$ so that the servo attempts to keep the arm still, the residual (7) splits between its inertial term $M\ddot{q}$ and its actuator term $h - \tau_{\text{act}}$ in a way that depends on the loop stiffness k_v . At $k_v = 1$ the arm is displaced 350 mm and 97.6 % of the residual is already in τ_{act} ; by $k_v = 20$ the displacement is 18 mm and the share is 100.0 % to four digits, with $\|M\ddot{q}\| = 1 \times 10^{-4}$ against $\|h - \tau_{\text{act}}\| = 1.314$; at $k_v = 100$ the displacement is 3.5 mm and the inertial term is numerically zero.

For any realistic loop gain the force appears almost entirely in τ_{act} . The servo rejects the disturbance, the arm barely moves, and the interaction is expressed as the torque the actuator must produce in order not to move. Two consequences follow, and they redirect the engineering effort in a way that is not obvious beforehand.

First, the finite-differenced acceleration, normally the delicate quantity in a residual observer, becomes irrelevant: it carries 0.006 % of the signal. Effort spent filtering encoder differences is misplaced on this platform.

Second, the entire error budget transfers to the current-to-torque calibration, where sensitivity is one to one: a 5 % error in the torque constant produces a 5 % error in force. A torque offset is milder, 0.05 N · m per joint producing 3.7 %.

B. DETECTION FLOOR

Combining Section V-A with (16) gives the smallest force distinguishable from residual error. A residual error of 0.01 N · m corresponds to 0.20 N at the median posture, 0.79 N at the fifth percentile of $\sigma_{\min}(J_v)$, and is unusable at the worst posture. Since the dominant term on a small geared servo is stiction, of order 0.05 N · m, the expected floor is one to two newtons at a typical posture. This is adequate for hand guiding with pushes of five to fifteen newtons, and it is the quantity that governs the design.

C. TWO COMPLIANCE REGIMES

Because the servo behaves as a velocity source, the arm does not yield under a push of its own accord, and compliance must be produced by displacing the reference: sense, move

TABLE 4. The two compliance regimes. Steady-state end-effector displacement, actual versus commanded by the admittance, under a push of increasing magnitude. The push is held for six admittance time constants and the commanded column matches the analytic equilibrium $\|f\| / (D_{\text{trans}}k)$ to 0.3 %. Every solve succeeded in every row.

$\ f\ $ [N]	commanded [mm]	actual [mm]	ratio	saturated steps
1	8.3	11.4	1.38	0 %
3	24.9	32.9	1.32	0 %
5	41.5	54.9	1.32	0 %
6	49.8	66.1	1.33	0 %
8	66.5	160.5	2.42	97 %
10	83.1	239.2	2.88	100 %

the reference, track. This synthetic compliance is limited by loop latency rather than by any physical property.

That description is only half correct, and the missing half is favorable. The actuator torque is bounded, so the force the arm can exert is bounded as well. Sweeping 729 configurations and evaluating the worst direction at each, subject to $|\tau_i| \leq 1.4 \text{ N} \cdot \text{m}$ and supporting the arm's own weight, the maximum exorable force has median 5.88 N, fifth percentile 4.29 N, and minimum 3.93 N. Manual guidance requires five to fifteen newtons. The interaction therefore has two regimes:

- *Below the capability of the posture, which is 4 to 6 N in the worst direction and higher in others.* The servo resists, the arm barely moves, and compliance is synthetic. Loop latency governs how the interaction feels.
- *Above that threshold.* The actuator saturates and the arm is displaced until it reaches a posture where it can hold the load. Compliance is physical and requires no control effort.

The transition between the two is not a matter of interpretation. It has a measurable signature: below the threshold the arm follows the reference the admittance commands, while above it the actuator saturates and the arm is carried further than commanded. Table 4 reports the ratio of actual to commanded end-effector displacement in the closed compliant loop under pushes of increasing magnitude.

The two indicators change together. Below the transition the arm follows the commanded displacement to within a constant factor of 1.32 to 1.38 and the actuator never saturates; above it the ratio rises to 2.42 and 2.88 and the actuator is saturated for 97 to 100 % of the push. The transition falls between 6 and 8 N, which is where it should fall: the static force this posture can exert in this direction is 6.53 N, computed independently from τ_{max} and the gravity load. The workspace figures quoted above, median 5.88 N and fifth percentile 4.29 N, are taken over the worst direction at each posture and are therefore lower than the capability of any single posture and direction; they bound the regime boundary from below rather than predicting it.

The arm is thus naturally backdrivable in the regime of interest, which is a safety property rather than a deficiency, and the synthetic loop only has to cover fine guidance and the return to task. Two practical consequences follow. The latency requirement on the control loop is milder than a purely

synthetic analysis would suggest, since above the threshold the physics yields without waiting for the loop. And actuator sizing becomes a compliance design variable: choosing a servo whose force capability falls below the expected interaction range confers intrinsic backdrivability with no control effort at all.

Remark 8 (The estimator does not saturate). *One might expect the estimate to saturate together with the actuator. It does not. Under a force ramp with $|\tau_i| \leq 1.4 \text{ N}\cdot\text{m}$ the estimate remains exact across the range, returning 1.000, 5.999, and 13.998 N for applied magnitudes of 1, 6, and 14 N, an error of 0.0 % throughout. The reason is that the residual (7) contains both terms: when the actuator can no longer hold, the arm accelerates and redistributes its posture, and the balance shifts from τ_{act} towards $\mathbf{M}\ddot{\mathbf{q}}$ and towards a configuration in which the load can be supported. The formulation captures both regimes without modification.*

VI. COMPLIANT CONTROL: AN INTEGRATION DEMONSTRATOR

A. ADMITTANCE

The estimated contact force drives a first-order admittance that displaces the reference pose from the nominal task pose, with a return spring that restores it when contact ceases,

$$\mathbf{D}\boldsymbol{\nu} = \hat{\mathbf{f}}_3 - k \mathbf{e}_{\text{task}}, \quad (17)$$

where $\boldsymbol{\nu}$ is the reference twist, $\mathbf{D}$ collects translational and rotational damping, k is the return stiffness, and $\mathbf{e}_{\text{task}}$ is the displacement of the reference from the task pose. The twist is integrated on the dual pose reference directly, so translation and rotation remain one object.

B. PREDICTIVE CONTROLLER

A nonlinear predictive controller tracks the compliant reference. The state is $\mathbf{x} = [\mathbf{q}^\top \dot{\mathbf{q}}^\top]^\top \in \mathbb{R}^8$, the input is joint torque, and the servo receives the predicted velocity $\dot{\mathbf{q}}_1^*$, which matches its native command mode. The optimal control problem is

$$\begin{aligned} \min_{\mathbf{x}, \boldsymbol{\tau}} \quad & \sum_{k=0}^{N-1} \|\mathbf{y}_k\|_{\mathbf{W}}^2 + \|\mathbf{y}_N\|_{\mathbf{W}_N}^2 \\ \text{s.t.} \quad & \mathbf{x}_0 = \hat{\mathbf{x}}, \quad \mathbf{x}_{k+1} = \Phi(\mathbf{x}_k, \boldsymbol{\tau}_k), \\ & \mathbf{q}_{\min} \leq \mathbf{q}_k \leq \mathbf{q}_{\max}, \quad \|\dot{\mathbf{q}}_k\|_{\infty} \leq \dot{\mathbf{q}}_{\max}, \\ & \|\boldsymbol{\tau}_k\|_{\infty} \leq \tau_{\max}, \end{aligned} \quad (18)$$

with residual $\mathbf{y} = [\mathbf{e}^\top \dot{\mathbf{q}}^\top \boldsymbol{\tau}^\top]^\top$ built on (6), and Φ the discretized dynamics (1) with $\mathbf{F} = \mathbf{0}$, since the interaction enters through the reference rather than the prediction model. The horizon is $N = 20$ nodes over 1.0 s.

C. MODEL REQUIREMENTS OF THE CONTROLLER

Formulation (18) takes torque as the decision variable while the actuator accepts velocity, so the predicted velocity is passed to the servo. This is the customary arrangement, but on velocity-actuated hardware it is not the only one, and the

TABLE 5. Controller formulations under velocity-level actuation. Regulation to a fixed pose from an initial error of 0.16 m, on the plant (5) with torque saturation.

	n_x	needs $\mathbf{M}, \mathbf{h}$	needs k_v	error [mm]	ms/solve
T	8	yes	no	0.180	2.41 / 3.08
V1	8	yes	yes	fails	2.41 / 3.14
V0	4	no	no	0.120	0.086 / 0.115

Error after settling; V0 with command weight tuned. V1 returns NAN_DETECTED at every solve, see text. Solve times are given for two runs of the same binary: the errors are deterministic and repeat exactly, the absolute timings do not, and the ratio between formulations does.

alternatives differ in what they require the designer to know. We implemented and compared three.

- **T**, torque level: $\mathbf{x} = [\mathbf{q}, \dot{\mathbf{q}}]$, $\mathbf{u} = \boldsymbol{\tau}$, the dynamics (1), with $\dot{\mathbf{q}}_1^*$ sent to the servo. Requires $\mathbf{M}$ and $\mathbf{h}$. Does not require k_v .
- **V1**, dynamic velocity level: $\mathbf{x} = [\mathbf{q}, \dot{\mathbf{q}}]$, $\mathbf{u}$ the velocity command, the plant (5). Requires $\mathbf{M}$, $\mathbf{h}$ and k_v , that is, identification of the servo loop.
- **V0**, kinematic: $\mathbf{x} = [\mathbf{q}]$, $\mathbf{u}$ the velocity command, $\dot{\mathbf{q}} = \mathbf{u}$, the limit of (5) as $k_v \rightarrow \infty$. Requires nothing: no inertia, no bias term, no servo gain.

Table 5 reports the comparison, and the outcome is not the expected one.

V1 is ill-posed. Every solve fails. The cause is instructive rather than numerical: with a proportional loop model, sustaining a velocity requires a persistent velocity error, so imposing the true actuator bound $|k_v(u_i - \dot{q}_i)| \leq \tau_{\max}$ restricts the command to $|u_i| \leq \tau_{\max}/k_v = 0.07 \text{ rad/s}$, which is 85 times below the 5.969 rad/s the servo actually reaches. Omitting the bound is worse, since the model then admits torques of 239 N·m, two orders of magnitude beyond the actuator, and the integrator diverges. The defect is the model structure: the real servo loop has integral action, so no persistent torque is needed to hold a velocity. Formulating V1 correctly therefore requires identifying not only a gain but the loop architecture including anti-windup, which is precisely the burden the other two formulations avoid.

V0 needs no model at all and is not worse. Once the command weight is tuned it settles to 0.120 mm against 0.180 mm for T, with no failures, and it solves between twenty-seven and twenty-eight times faster. We quote the ratio rather than the absolute time because the ratio is what repeats: across two runs of the same binary the settled errors are identical to the digit while the absolute solve times move by a third. Its state is half the size and it uses neither the inertia matrix nor the bias term.

This yields a clean separation of concerns, which we regard as the practical recommendation of this paper:

On a velocity-actuated arm the dynamic model is required by the estimator and by nothing else. The controller can be model-free.

The estimator needs $\mathbf{M}$ and $\mathbf{h}$ to form the residual (7), and by Remark 1 it needs no knowledge of the servo loop. The

controller needs neither. The only quantities to be identified on hardware are therefore the kinematics, the mass properties, and the current-to-torque constant, all of which serve estimation, and none of which serve control.

Remark 9 (Limitations of V0). *The kinematic formulation cannot express torque bounds, so it cannot by itself prevent commands the actuator will not follow. Two considerations make this acceptable here. The velocity bound is retained and is the binding limit in normal operation, and by Section V-C exceeding the torque bound does not produce a fault but the physically compliant regime, in which the arm yields. If a hard torque bound is required, for instance to protect a payload, the static requirement $\|\mathbf{h}(\mathbf{q}) + \mathbf{J}_v^\top \hat{\mathbf{f}}\|_\infty \leq \tau_{\max}$ can be imposed on V0 as a state constraint without reintroducing k_v .*

Table 5 compares the three under regulation. The claim that the controller can be model-free is stronger than a regulation test, however, so Section VII-C repeats the complete compliant interaction loop with V0 in place of T.

Problem (18) is transcribed with CasADi [28] and solved by the real-time iteration scheme [26] in acados [27], with HPIPM [29], a Gauss-Newton Hessian, and a fourth-order explicit Runge-Kutta integrator.

Because $n = 4$, a general SE(3) reference is unreachable and the optimizer settles on a compromise between orientation and translation error. Which compromise it selects is decided by the metric, which is what $\mathbf{J}_\ell^{-1}$ in (6) supplies.

Remark 10 (Symbolic form decides the solve rate). *Writing $\ddot{\mathbf{q}} = \text{solve}(\mathbf{M}, \boldsymbol{\tau} - \mathbf{h})$ on CasADi's graph-valued type emits a node backed by a numerical factorization rather than a closed-form expression, differentiated implicitly and evaluated some eighty times per solve by the integrator: the cost was 101.6 ms, of which 101.5 ms was linearization and 0.13 ms the quadratic program, so the optimizer was never the bottleneck. Rebuilding the identical model on the scalar type turns that solve into symbolic Gaussian elimination and gives 2.53 to 2.97 ms mean over 1949 consecutive solves, with the controller unchanged and the two builds agreeing to 2.2×10^{-15} N · m in commanded torque over 250 closed-loop steps. The failure mode presents as a slow solver rather than as a modeling error, which is why it is recorded here.*

VII. RESULTS

Estimation is reported first against an analytic plant, where ground truth is exact, and then against MuJoCo, which provides an independent implementation of the dynamics including effects absent from (1).

A. FORCE ESTIMATION IN CLOSED LOOP

A hand-guiding scenario is simulated at 100 Hz. The arm holds a task pose, an external force follows a trapezoidal profile, and the compliant loop runs throughout. Both estimators are computed from the same residual at every step, so the only difference is the assumed contact model. Two push directions

TABLE 6. Closed-loop estimation from an identical residual under two contact models.

push	estimator	RMSE [N]	corr.	error
in plane	six-dim.	0.024	1.0000	0.1 %
$\ \mathbf{f}\ = 6.325$ N	point	0.008	1.0000	0.0 %
normal	six-dim.	4.832	0.8069	83.4 %
$\ \mathbf{f}\ = 6.000$ N	point	0.007	1.0000	0.0 %

are used: one in the plane of the arm and one normal to it, the latter being the direction identified in Section IV-F.

Table 6 and Fig. 4 report the outcome. For the in-plane push both estimators are accurate, which is the regime in which the conventional inversion has historically been validated. For the normal push the six-dimensional inversion is wrong by 83.4 % while the point-contact inversion remains exact to the discretization floor of the benchmark. In both cases the compliant loop yields under the push and returns on release, with the solver reporting success at all 350 steps.

The same result holds without reference to any angle. For three directions that name themselves, evaluated at four configurations, a push along the arm or along the base axis lies in the plane of the arm and the six-dimensional inversion recovers between 97.3 and 100.0 % of it, whereas a push normal to that plane yields only 6.7 to 23.2 % and a spurious moment of 0.04 to 0.20 N · m per newton applied. The point-contact inversion returns 100.0 % in every one of those twelve cases.

B. VALIDATION AGAINST AN INDEPENDENT ENGINE

Replacing the analytic plant with MuJoCo and applying a known external force gives a check that does not share the implementation of the dynamics. At the push peak the applied force is [0.00, 4.00, −1.50] N; the six-dimensional inversion returns [0.59, 0.23, −0.83] N and the point-contact inversion [0.21, 3.92, −1.31] N. Over the push window the RMSE against the applied force falls from 2.227 N to 0.157 N, a factor of fourteen.

Remark 11 (Interpretation of the residual error). *The remaining 0.157 N is not estimator error, since the same estimator is exact to 10^{-15} on the analytic plant. It is the model-to-engine gap: joint damping and friction, the discrete constraint solver, and effects not represented in (1). It is the best available proxy for hardware degradation and should be read as a lower bound.*

Remark 12 (Correlation is not accuracy). *In the same experiment the six-dimensional estimate correlates with ground truth at 0.997 while carrying the 2.227 N error. That correlation measures association rather than agreement, and is insensitive to gain and offset, is a long-standing point in measurement comparison [33]; what the experiment adds is that the failure mode is reached here by the estimator's own structure rather than by noise, so a validation of this kind can appear sound while the quantity of interest is substantially*

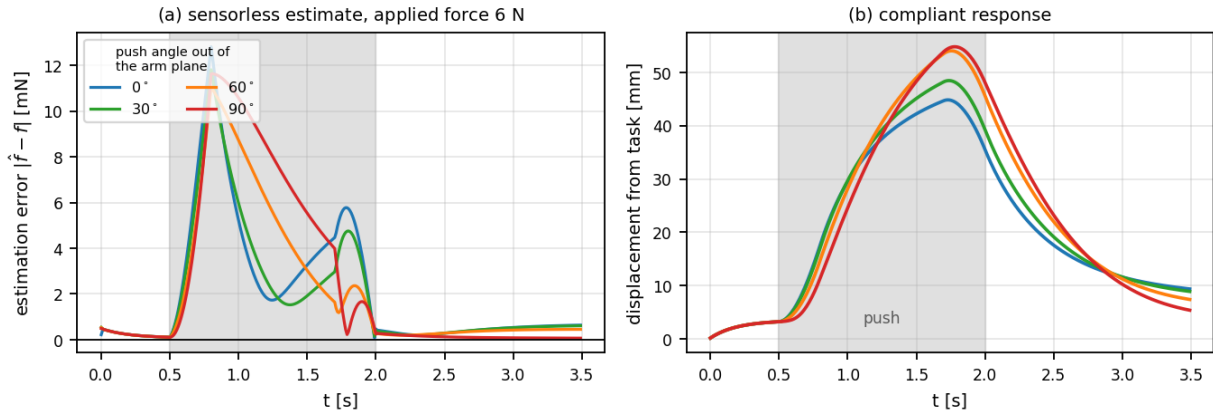


FIGURE 4. Closed-loop interaction under a 6 N push applied at four angles out of the plane of the arm. (a) Error of the point-contact estimate. It stays below 13 mN, that is under 0.25 % of the applied force, at every angle including the one the six-dimensional inversion cannot recover; the residual excursions coincide with the ramps of the force profile and are the discretization floor of the benchmark rather than a directional effect. (b) The resulting compliant behavior: the arm yields by 45 to 55 mm under the push and returns towards the task pose on release, at every angle.

TABLE 7. Complete compliant loop with a model-based controller (T) and a model-free controller (V0). The estimator is identical in both.

push	ctrl	force RMSE [N]	yield [mm]	return [mm]	ms/solve
in plane	T	0.0299	127.1	11.23	2.98
	V0	0.0332	130.7	9.24	0.119
normal	T	0.0042	23.0	3.71	3.05
	V0	0.0045	28.7	1.92	0.123

wrong.

C. THE COMPLIANT LOOP WITHOUT A DYNAMIC MODEL

Section VI-C argued from a regulation test that the controller need not carry a dynamic model. We repeat here the complete interaction loop of Section VII-A, identical in every respect except that the kinematic controller V0 replaces the torque-level controller T. The estimator is unchanged and retains the dynamic model, which it requires.

Table 7 reports the outcome. Force estimation is unaffected, as it must be, since the estimator did not change. The compliant behavior is equivalent: the arm yields comparably under the push and returns closer to the task pose on release, in both push directions. The computational cost falls by a factor of twenty-five. Carrying the inertia matrix and the bias term in the controller therefore brings no benefit on this platform, while carrying them in the estimator is unavoidable.

D. ROBUSTNESS TO SERVO NOISE

Every result reported so far is noise-free, and on a clean bench an exact estimator looks perfect by construction. We therefore evaluate both inversions under a noise model built from the published specifications of the MX-28R, not from measurements of one: current quantization at 3.36 mA per count, Coulomb stiction of 0.05 N · m per joint with random sign, which Section VII-F identifies as the dominant term, and twelve-bit encoder quantization. The stiction magnitude

TABLE 8. Monte Carlo under servo noise, $N = 2000$. Relative force error, as a percentage of the applied magnitude.

noise	six-dimensional			point contact		
	median	p90	p99	median	p90	p99
none	47.2	86.8	97.9	0.0	0.0	0.0
nominal	69.6	149.1	1623	18.0	58.9	353
$\times 2$	89.3	281.2	3194	35.9	118.3	695
$\times 4$	137.3	548.4	6239	71.9	232.8	1402

TABLE 9. Effect of rejecting ill-conditioned postures, nominal noise.

$\sigma_{\min}(\mathbf{J}_v)$ threshold	workspace kept	six-dimensional		point contact	
		median	p90	median	p90
0.00	100.0 %	69.6	149.1	18.0	58.9
0.02	88.6 %	70.3	151.8	16.5	42.1
0.04	68.7 %	69.7	144.1	14.2	36.1
0.06	31.4 %	77.6	225.8	12.3	29.1

in particular is an assumption rather than a datasheet figure, so this experiment bounds how the two inversions respond to noise of that character and does not predict the error the physical servo will produce. Configurations are drawn uniformly over the joint ranges and contact forces uniformly between 2 and 10 N, over 2000 trials per condition.

Table 8 reports the outcome, and it is the honest expectation for hardware. The point-contact inversion is no longer exact: its median relative error is 18.0 % under nominal noise, against 69.6 % for the six-dimensional one. The advantage is preserved at every percentile and under degraded noise, but both distributions have heavy tails, which the conditioning analysis of Section IV predicts and the next experiment confirms.

Table 9 discriminates between the two error mechanisms. Rejecting postures below a threshold on $\sigma_{\min}(\mathbf{J}_v)$ halves the ninetieth percentile of the point-contact estimator, from 58.9 % to 29.1 %, exactly as (16) predicts for an error that

is amplified measurement noise. It does nothing for the six-dimensional inversion, whose error is not noise but structural misattribution and therefore cannot be gated away. The two mechanisms identified analytically in Section IV are thus separable experimentally, and monitoring $\sigma_{\min}(\mathbf{J}_v)$ online becomes an actionable recommendation with a measured benefit.

The benefit is not free, and the second column states its price. The threshold that halves the tail accepts only 31.4 % of the workspace; at $\sigma_{\min}(\mathbf{J}_v) = 0.04$ the arm keeps 68.7 % of the workspace and still gains a third of the tail reduction. Which point on that trade-off is right depends on the task, and reporting the error percentile without the retained fraction would make an expensive gate look like a free improvement. The gate is also better used as a confidence flag on a reported force than as a constraint on where the arm may go, since a four-joint arm tracking a pose has no redundancy with which to satisfy it. Both tables are computed on 2000 samples from the same seed, so the 0.00 row reproduces the nominal-noise row of Table 8 exactly.

Since the entries are sample statistics, their run-to-run spread bounds what may be read into them. Repeating the nominal case over five seeds gives a ninetieth-percentile point-contact error in [58.9, 62.0] % and, under the 0.06 gate, in [28.4, 29.8] %; the intervals are disjoint by a wide margin, so the halving is not a sampling artifact. The six-dimensional median moves from [68.5, 72.8] % to [73.2, 77.6] % under the same gate, confirming that gating does not help it. Differences smaller than a few percentage points in these tables should not be interpreted.

E. DIRECT VERIFICATION OF THE CONDITIONING BOUND

Section VII-D showed that rejecting ill-conditioned postures shortens the tail of the point-contact estimator, which is consistent with (16) but does not measure its exponent. We test the bound directly and through the independent plant.

The arm is held at four postures spanning $\sigma_{\min}(\mathbf{J}_v)$ from 0.011 to 0.092, a factor of nine, selected to be free of contact, clear of the floor, and away from the joint limits. Sensor noise is injected on the current reading, which is where the servo carries it, at one and at three quantization steps of the MX-28R. For each posture we estimate the force sample by sample and record the dispersion of the estimate, which is the quantity (16) bounds.

Table 10 reports the outcome. Fitting $\log \text{std}(\hat{\mathbf{f}})$ against $\log \sigma_{\min}(\mathbf{J}_v)$ gives a slope of -0.872 with $R^2 = 0.980$ at the lower noise level and -0.790 with $R^2 = 0.987$ at the higher one, against the predicted -1 . The dispersion therefore scales essentially inversely with $\sigma_{\min}(\mathbf{J}_v)$ across a ninefold range, at two noise levels, and through a plant whose damping, armature, and dry friction are absent from the estimator model.

Two details are worth stating. The measured dispersion is consistently below the bound, at 0.48 to 0.77 of it, which is as it should be: (16) is an upper bound attained only when the residual error aligns with the least observable direction.

TABLE 10. Noise amplification through the MuJoCo plant. Dispersion of the point-contact estimate at four postures, with sensor noise injected on the current reading.

$\sigma_{\min}(\mathbf{J}_v)$	$\text{std}(\boldsymbol{\tau})$	$\text{std}(\hat{\mathbf{f}})$	bound (16)	ratio
<i>current noise $\sigma = 3.4$ mA (one quantization step)</i>				
0.0108	0.0039	0.183	0.361	0.51
0.0204	0.0036	0.096	0.175	0.55
0.0435	0.0037	0.044	0.085	0.51
0.0918	0.0035	0.030	0.038	0.77
<i>current noise $\sigma = 10$ mA (three quantization steps)</i>				
0.0108	0.0102	0.451	0.945	0.48
0.0204	0.0103	0.250	0.504	0.50
0.0435	0.0102	0.128	0.233	0.55
0.0918	0.0102	0.085	0.111	0.76

TABLE 11. Spurious force reported under no contact when a single model or calibration parameter is perturbed, at the most sensitive of six configurations.

perturbation	spurious $\ \hat{\mathbf{f}}_3\ $ [N]
inverted torque sign	8.75
torque constant $\times 2$	4.38
torque offset 0.05 N · m	1.31
stiction 0.05 N · m	1.09
torque constant +20 %	0.88
torque offset 0.02 N · m	0.52
link 4 mass +20 %	0.50
joint zero shifted 0.10 rad	0.33
<i>acceptance threshold 0.3 N</i>	
link 3 mass +20 %	0.29
torque constant +5 %	0.22
joint zero shifted 0.05 rad	0.17
link 2 mass +20 %	0.04
link 1 mass +20 %	0.00

And the fitted slope is slightly shallower than -1 for the same reason, since the alignment varies with posture.

Remark 13. *The noise here is modeled rather than measured: MuJoCo is deterministic, so at equilibrium the actuator torque is exactly constant and quantization alone produces zero variance. What this experiment establishes is that the bound governs the estimator when it is driven through the full nonlinear plant, not merely in the algebraic setting of Section IV. Replacing the modeled noise with a real servo's is the remaining step, and it requires only a static measurement with the actuators disabled.*

F. SENSITIVITY TO MODEL AND CALIBRATION ERROR

Section V places the error budget on the current-to-torque calibration, and Section VII-D shows stiction dominating the noise. Both statements can be made precise by perturbing the validated model one parameter at a time and recording the spurious force the estimator then reports under no contact, which is the quantity a deployment would have to distinguish from a real interaction. Table 11 reports it at the worst of six configurations selected by sweeping the workspace for maximum sensitivity.

Two features of Table 11 matter for deployment. Torque offsets dominate: 0.02 N·m already produces 0.52 N, whereas a 20 % error in a link mass produces 0.50 N and a 5 % error in the torque constant only 0.22 N. The asymmetry is geometric. A gravity error lies in the well-conditioned directions of $\mathbf{J}_v$ and is largely absorbed by the least-squares projection of (15), whereas an offset is not, which is why stiction rather than mass modeling sets the floor. The estimate is also insensitive to the mass of the first link, correctly so, since that link rotates about a vertical axis and gravity exerts no torque about it.

A practical consequence follows for calibration. Because gravity torques on this arm are of order 0.3 N·m, a no-contact null test cannot resolve the torque constant: a 5 % error in it produces 0.22 N, below the threshold and indistinguishable from the residual bias of a well-modeled arm. The constant must therefore be identified under a known load rather than at rest.

G. TRACKING PERFORMANCE OF THE DEMONSTRATOR

The integration demonstrator behaves as required of it and no more is claimed. Regulating to a fixed pose from an initial error of 0.160 m converges to 1.65 mm with success at all 250 steps, and on a slow screw trajectory it achieves 6.82 mm position RMSE (maximum 7.40 mm) and 4.79° orientation RMSE (maximum 6.35°). Absolute solve times are quoted as ratios throughout, since two runs of an identical build differ by 17 % in the mean and 59 % in the maximum, whereas the settled errors repeat to the digit.

VIII. DISCUSSION

A. PRACTICAL IMPLICATIONS

The correction of Section IV is inexpensive to adopt. It replaces one linear solve with another of smaller dimension and requires no additional sensing, calibration, or tuning. For any application in which the contact location is known by construction, and hand guiding at the end-effector is the archetype, the six-dimensional inversion should not be used on an arm with fewer than six joints.

The analysis also reframes the platform. A four-joint arm sensing contact force through its own actuators is not operating at the margin of feasibility: by Proposition 4 it has one measurement more than the problem requires. The quantity a designer should examine when placing a task is $\sigma_{\min}(\mathbf{J}_v)$, mapped in Fig. 2(b), not any index derived from the rank deficit of the full Jacobian. Velocity-level actuation, in turn, removes the need to identify the servo loop and renders the acceleration term negligible, concentrating the design effort on a single calibration.

B. LIMITATIONS

Simulation only. Every number reported here is obtained in simulation. The model agrees with MuJoCo to 10^{-10} and MuJoCo serves as an independently written plant, which establishes that two implementations of the same rigid-body mechanics coincide and nothing beyond that; neither substitutes for the physical arm, and the agreement figure should

not be read as evidence of physical fidelity. Two quantities are expected to degrade on hardware: the actuator-torque proxy from servo current, which carries stiction and gearbox losses absent from (1), and the finite-differenced acceleration, which will be noisier, although Section V shows the latter matters little. The 0.157 N model-to-engine gap is the best available proxy and is a lower bound.

The magnitude of the misattribution is metric-dependent. By Propositions 2 and 3 the directions the six-dimensional inversion loses, and the forces it recovers exactly, are properties of the configuration; how much it loses is not, and depends on the length ℓ_c that renders the wrench norm homogeneous. Table 2 reports the dependence over a plausible range, within which the mean discarded fraction spans 12.8 to 51.4 % while the worst case stays at 99.6 %. Any single figure quoted from this paper, the 50.5 % included, is therefore a statement about the arm *and* about the unweighted convention it was computed under. A designer applying the analysis to another arm should recompute Table 2 rather than transport the percentage, and should declare ℓ_c and the point about which the moment is taken. What transports without recomputation is the geometric argument, the directional structure, and Proposition 4.

Known contact point. As stated in Remark 5, the point-contact model requires the contact location. Contact along the links, or contact transmitting a moment through a grasped object, lies outside the analysis.

Conditioning near singularities. Although $\text{rank } \mathbf{J}_v = 3$ throughout the sampled workspace, $\sigma_{\min}(\mathbf{J}_v)$ reaches 7.6×10^{-5} , where (16) predicts severe amplification. A deployment should monitor $\sigma_{\min}(\mathbf{J}_v)$ online, which is inexpensive, and either avoid those postures or widen the admittance dead band there.

Friction is not observed by either route. The servo current measures motor torque upstream of the gearbox, so gearbox friction is unmodeled whether the applied torque is read from current or inferred from the velocity loop. This sets the irreducible floor and is the quantity most likely to dominate on hardware.

Statistical scope. Every result rests on a single simulator, and the timing figures vary by 17 % in the mean between two runs of an identical build, which is why solve times are quoted as ratios and reported for both runs rather than for the more favorable one.

C. PROPOSED HARDWARE VALIDATION

Two of the claims above are falsifiable on the physical arm at negligible cost, and neither requires a force/torque sensor, a control loop, or a fast bus. The arm is held with actuator torque disabled and positioned by hand at the four postures of Fig. 2(b), spanning a factor of nine in $\sigma_{\min}(\mathbf{J}_v)$.

The first measures amplification. With nothing applied, the sample-to-sample dispersion of $\hat{\mathbf{f}}_3$ is recorded at each posture; (16) predicts that $\log \text{std } \hat{\mathbf{f}}_3$ regresses on $\log \sigma_{\min}(\mathbf{J}_v)$ with unit negative slope. A slope near zero refutes Sec-

tion IV-E. This part cannot be rehearsed in simulation, since a deterministic engine has no dispersion to measure.

The second measures accuracy and, more importantly, misattribution. A known load is applied at the end-effector and both inversions are evaluated on the same residual. A hanging mass supplies a vertical force and a luggage scale supplies a horizontal one; two directions are required because a vertical force exerts no moment about the vertical base axis and would leave that joint unexcited. The prediction of Proposition 1 is that the six-dimensional inversion returns a nonzero moment where the true one is zero, and the outcome that would refute this paper is the two inversions agreeing. The comparison is differential, so the two measurements must share a Jacobian: if the arm sags under the load, the posture change contaminates the difference and the test returns no verdict.

IX. CONCLUSION

This paper turns an established dimensional argument into quantities a designer can use. On a 4-DoF arm the six-dimensional wrench inversion discards 50.5 % of an applied unit force on average over 2197 configurations, and the force block of the wrench projector predicts which directions are lost. The magnitude depends on the characteristic length that renders the wrench norm homogeneous, falling to 38.0 % at the arm's own scale, while the directions and the set of exactly recovered forces do not. The failure is also silent, coexisting with a 0.997 correlation against ground truth. The point-contact inversion recovers the force wherever rank $\mathbf{J}_v = 3$, which holds throughout the sampled workspace, leaving conditioning as the binding limitation and $\sigma_{\min}(\mathbf{J}_v)$ as the quantity to monitor.

Velocity-level actuation, commonly regarded as an obstacle to sensorless estimation, is instead a simplification: the servo reports the torque it applies, so the residual is invariant to the inner loop, and a kinematic controller matches a model-based one throughout the compliant loop at one twenty-fifth of the cost. Because the actuator saturates below the force range of manual guidance, physical compliance requires no control effort.

The study is limited to simulation, and the estimator has not been shown to survive gearbox friction, stiction, backlash, or the calibration of the torque constant. Future work is the hardware protocol of Section VIII-C, which tests the directional prediction, the conditioning law, and the viability of inferring applied torque from servo current, and extending the contact model to an unknown location along the links.

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